13.9.1 Photolysis of Water

The process of Photolysis is associated with **Oxygen Evolving Complex** (OEC) or water splitting complex in pigment system II and is catalysed by the presence of Mn^{++} and Cl^{-}. When the pigment system II is active it receives light and the water molecule splits into OH^{-} ions and H^{+} ions. The OH^{-}ions unite to form water molecules again and release O_2 and electrons (Figure 13.11).

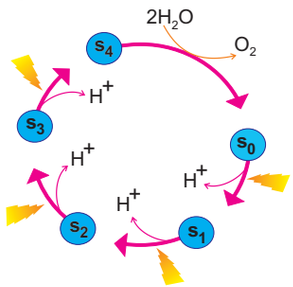


Figure 13.11: Oxygen Evolving Complex (OEC)

$$4\ H_2O \longrightarrow 4\ H^+ + 4\ OH^-$$
$$4\ OH^- \longrightarrow 2\ H_2O + O_2 + 4\ e^-$$
$$2H_2O \longrightarrow 4\ H^+ + O_2 + 4\ e^-$$

13.9.2 Electron Transport Chain of Chloroplast

Electron Transport Chain in each photosystem involves four complexes:

- *Core Complex (CC):* CC I in PS I the reaction centre is P700, CC II in PS II the reaction centre is P680
- Light Harvesting Complex or Antenna complex (LHC):
- Two types: LHC I in PS I and LHC II in PS II.
- Cytochrome b6 f complex: It is the non-pigmented protein complex connecting PS I and PS II. Plastoquinone (PQ) and Plastocyanin (PC) are intermediate complexes acting as mobile or shuttle electron carriers of Electron Transport Chain. PQ acts as shuttle between PS II and Cytochrome b_6- f complex and PC connects
- Cytochrome b_6-f and PS I complex.
- *ATPase complex or Coupling factor:* It is found in the surface of thylakoid membrane. This complex is made up of CF_1 and CF_0 factors. This complex utilizes energy from ETC and converts ADP and inorganic phosphate (P_i) into ATP (Figure 13.12).

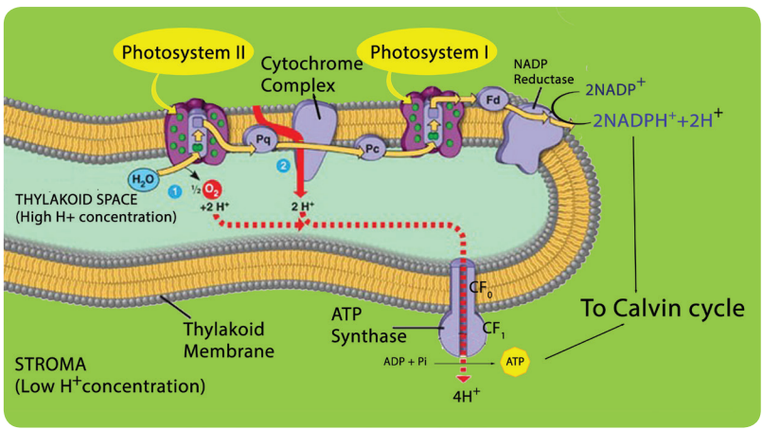


Figure 13.12: Electron Transport Chain in Chloroplast

13.10 Photophosphorylation

Phosphorylation taking place during respiration is called as **oxidative phosphorylation** and ATP produced by the breakdown of substrate is known as **substrate level phosphorylation**. In this topic, we are going to learn about phosphorylation taking place in chloroplast with the help of light. During the movement of electrons through carrier molecules ATP and NADPH + H^+ are produced. Phosphorylation is the process of synthesis of ATP by the addition of inorganic phosphate to ADP. The addition of phosphate here takes place with the help of light generated electron and so it is called as **photophosphorylation**. It takes place in both cyclic and non-cyclic electron transport.

13.10.1 Cyclic Photophosphorylation

Cyclic photophosphorylation refers to the electrons ejected from the pigment system I (Photosystem I) and again cycled back to the PS I. When the photons activate P700 reaction centre photosystem II is activated. Electrons are raised to the high energy level. The primary electron acceptor is Ferredoxin Reducing Substance (FRS) which transfers electrons to Ferredoxin

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